WORKED SOLUTION

STEP II 2001, Question 8

Pure • Periodic Functions and ODEs

STEP II 2001, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s2-q8

The key idea

A function with period 1 looks the same on every interval of length 1, so its integral over any such interval is the same number. That makes F grow by the same amount, $F(1)$, every time x goes up by 1. The differential equation in the last part is first order and linear, so an integrating factor turns its solution into an exponential of $-F(x)$, and the rest follows from the fact that $F(1)$ is positive.

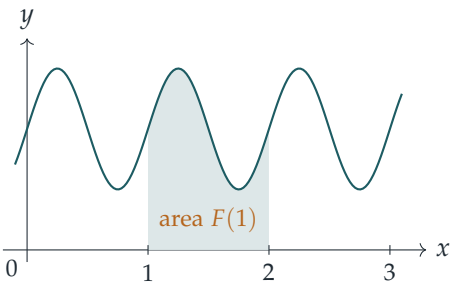
Solution

Part (i)

Take

$$f(x) = 2 + \sin(2\pi x).$$

Since sin has period 2π, $\sin(2\pi(x+1)) = \sin(2\pi x + 2\pi) = \sin(2\pi x)$, so $f(x+1) = f(x)$. Also $\sin(2\pi x) \geq -1$, so $f(x) \geq 1 > 0$ for every x. (A constant such as $f(x) = 1$ also works, but is less interesting.)



Every unit-width strip under the graph has the same area; that is the whole of part (ii).

Part (ii)

Define $G(x) = F(x+1) - F(x)$. Differentiating, using the chain rule on the first term,

$$G'(x) = f(x+1) - f(x) = 0$$

for all x, by the periodicity of f. So G is constant, and its value is $G(0) = F(1) - F(0) = F(1)$, since $F(0) = 0$. That is,

$$F(x+1) = F(x) + F(1) \quad \text{for all } x.$$

Now we use induction on n. For $n = 1$ the claim $F(1) = 1 \cdot F(1)$ is trivially true. If $F(k) = kF(1)$ for some positive integer k, then putting $x = k$ above,

$$F(k+1) = F(k) + F(1) = kF(1) + F(1) = (k+1)F(1).$$

Hence $F(n) = nF(1)$ for every positive integer n, as required.

(Equivalently, if f is continuous, $F(x) = \int_0^x f(t)\,\mathrm{d}t$ and $F(n)$ is the sum of n integrals over $[k, k+1]$, each equal to $\int_0^1 f(t)\,\mathrm{d}t$ after the substitution $t \mapsto t + k$. The argument above avoids having to write F as an integral at all.)

Part (iii)

The equation is linear in y, and the integrating factor is $\mathrm{e}^{\int f\,\mathrm{d}x} = \mathrm{e}^{F(x)}$, which is where F comes in. Indeed, by the product rule and $F' = f$,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(y\,\mathrm{e}^{F(x)}\right) = \mathrm{e}^{F(x)}\left(\frac{\mathrm{d}y}{\mathrm{d}x} + f(x)\,y\right) = 0.$$

So $y\,\mathrm{e}^{F(x)}$ is constant. At $x = 0$ it equals $1 \cdot \mathrm{e}^{F(0)} = 1$, so

$$y = \mathrm{e}^{-F(x)}.$$

By part (ii), for each positive integer n,

$$y(n) = \mathrm{e}^{-F(n)} = \mathrm{e}^{-nF(1)} = \left(\mathrm{e}^{-F(1)}\right)^n.$$

Ask yourself How do we know $F(1) > 0$? We are not told anything about $F(1)$ directly.

Since $F'(x) = f(x) > 0$ for all x, F is strictly increasing, and so $F(1) > F(0) = 0$. (If you prefer integrals: $F(1)$ is the area under a positive curve between 0 and 1.)

Therefore $0 < \mathrm{e}^{-F(1)} < 1$, and a fixed number r with $0 < r < 1$ satisfies $r^n \to 0$ as $n \to \infty$. Hence $y(n) \to 0$ as $n \to \infty$, which is what we had to show.

Notice that we only claim this along the integers. Between integers, y is squeezed: since F is increasing, y is decreasing, so in fact $y(x) \to 0$ as $x \to \infty$ through all real values too.

Checks

With $f(x) = 2 + \sin(2\pi x)$ we have $F(1) = \int_0^1 (2 + \sin 2\pi t)\,\mathrm{d}t = 2$, since the sine integrates to zero over a full period. Simpson's rule gives $F(2), F(3), F(5), F(7)$ equal to $4, 6, 10, 14$ to better than 10^{-12}. Stepping the differential equation numerically (fourth-order Runge–Kutta) from $y(0) = 1$ gives $y(1) \approx 0.135335 = \mathrm{e}^{-2}$ and $y(4) \approx 3.3546 \times 10^{-4} = \mathrm{e}^{-8}$, matching $\mathrm{e}^{-nF(1)}$. We also checked $F(x+1) - F(x) = F(1)$ at non-integer x for the less symmetric example $f(x) = \mathrm{e}^{\cos 2\pi x} + 0.1$.

Common slips

- Writing $F(n) = nF(1)$ as "obvious" from periodicity without showing why the area over $[k, k+1]$ equals the area over $[0, 1]$. The function $G(x) = F(x+1) - F(x)$ or a substitution is needed.
- Forgetting to justify $F(1) > 0$. Without it, $\mathrm{e}^{-nF(1)}$ could stay at 1 or blow up.
- Getting the sign of the integrating factor wrong and ending with $y = \mathrm{e}^{F(x)}$, which grows instead of decaying.

Beyond STEP

This is the simplest case of *Floquet theory*: for a linear differential equation with periodic coefficients, the solution after one period is the starting value multiplied by a fixed factor (here $\mathrm{e}^{-F(1)}$), so the long-term behaviour is decided by whether that factor is less than or greater than 1. The same idea, with matrices in place of numbers, explains the stability of a pendulum whose support is shaken up and down. Boyce and DiPrima, *Elementary Differential Equations*, covers linear equations and integrating factors thoroughly and is a good next step.

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